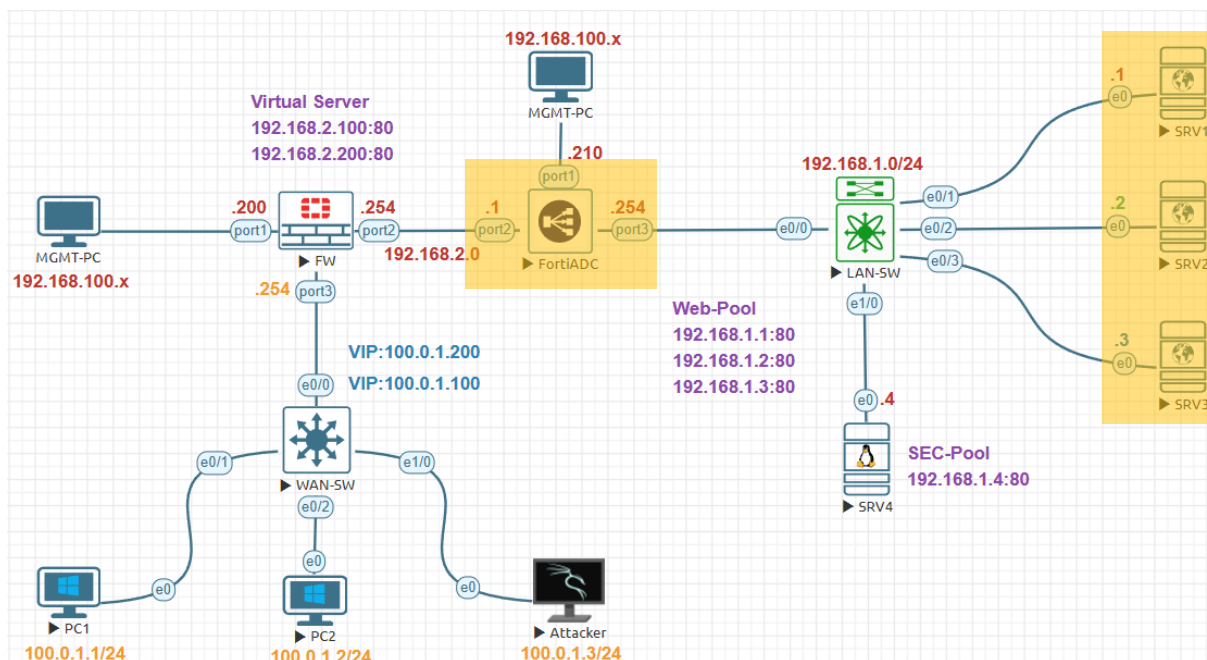


HTTP Health Check Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Let's create our own custom Health Check Navigate to Shared **Resources>Health Check>Health Check** click on **Create New**

FortiADC FortiADC

Dashboard > Health Check Health Check Monitor Health Check Script

Security Fabric > Delete Create New Add Filter

FortiView >

System >

Shared Resources >

Health Check

Name	Type
LB_HLTHCK_ICMP	ICMP
LB_HLTHCK_HTTP	HTTP
LB_HLTHCK_HTTPS	HTTPS
LB_HLTHCK_TCP_ECHO	TCP Echo

Showing 1 to 4 of 4 entries 0 rows selected Show 25 entries

Network >

Type **Name** in this case **HTTP-HealthCheck** and choose Type **HTTP** click **Save**.

Health Check

Dashboard >

Security Fabric >

FortiView >

System >

Shared Resources >

Health Check

Schedule Group

Address

Service

Network >

Server Load Balance >

Link Load Balance >

Global Load Balance >

Web Application Firewall >

Network Security >

DoS Protection >

User Authentication >

Log & Report >

Name HTTP-HealthCheck

Type HTTP

Specifics

Port 80
Range: 0-65535

Http Connect No Connect Local Connect Remote Connect

Method Type HTTP Get HTTP Head

HTTP Version HTTP 1.0 HTTP 1.1

Send String /index.html

Receive String Optional. Specify the string.

Status Code 200

Match Type Match String Match Status Match All

Username Optional. Specify the username.

Password Specify the password, if any.

Hostname Optional. Specify the host name.

Additional String Optional. Specify the additional string.
The non-empty additional string should end with \r\n.

Next navigate to **Server Load Balance > Real Server Pool > Real Server Pool** tab double click on **WEB-Pool** to open the setting.

Real Server Pool Real Server Server SSL

Delete + Create New + Add Filter

Name	Address Type	Health Check
WEB-Pool	IPv4	Enable
SEC-Pool	IPv4	Disable

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

Server Load Balance

- Virtual Server
- Application Resources
- Application Optimization
- Real Server Pool**

Toggle to enable the **Health Check** on right side Double-click to choose **HTTP-HealthCheck** previously created click **Save** to apply the Health Check it will inherited automatically.

Real Server Pool

Name: WEB-Pool

Address Type: IPv4 IPv6

Type: Static Dynamic

Health Check: ☒

Health Check Relationship: AND OR

Health Check List: HTTP-HealthCheck

Available Items: LB_HLTHCK_HTTP, LB_HLTHCK_HTTPS, LB_HLTHCK_TCP_ECHO, ICMP-HealthCheck, TCP-HealthCheck

Direct Route Mode: ☐

Real Server SSL Profile: NONE

Member

ID	Name	Address	Health Check
1	SRV1	192.168.1.1	inherited
2	SRV2	192.168.1.2	inherited
3	SRV3	192.168.1.3	inherited

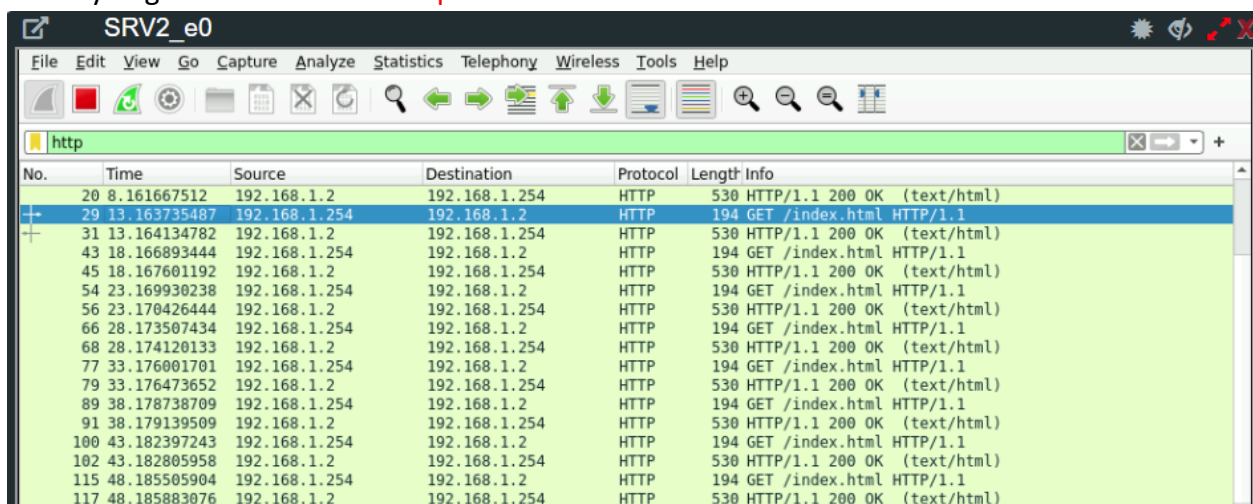
Save Cancel

Verification:

Go to SRV1 login with username & password root/root type **tcpdump tcp**.

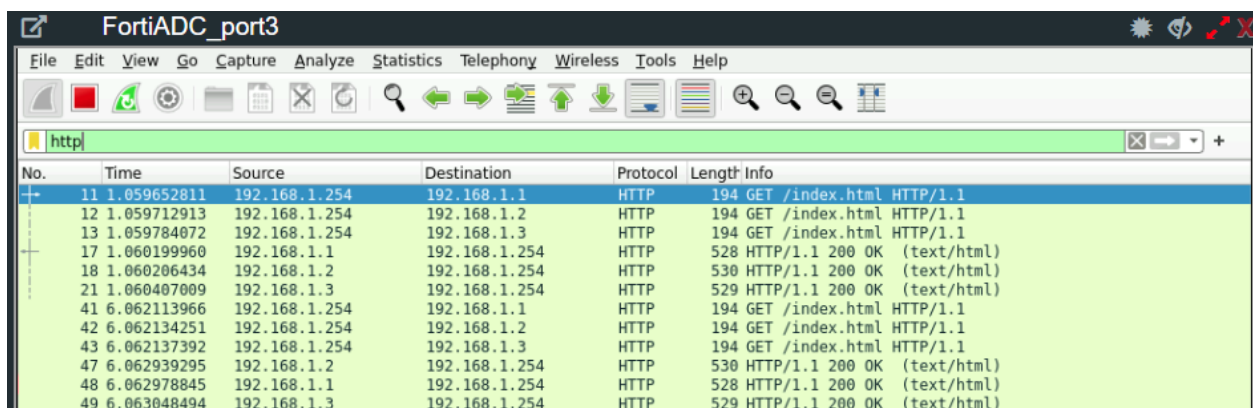
```
SRV1 x
root@SRV1:~#
root@SRV1:~# tcpdump tcp
tcpdump: verbose output suppressed, use -v or -vv for full
listening on ens3, link-type EN10MB (Ethernet), capture size
08:40:28.053523 IP 192.168.1.254.21014 > 192.168.1.1.http:
8101 ecr 0,nop,wscale 9], length 0
08:40:28.053599 IP 192.168.1.1.http > 192.168.1.254.21014:
OK,TS val 62293 ecr 1788788101,nop,wscale 7], length 0
08:40:28.054185 IP 192.168.1.254.21014 > 192.168.1.1.http:
length 0
08:40:28.054542 IP 192.168.1.254.21014 > 192.168.1.1.http:
r 62293], length 128: HTTP: GET /index.html HTTP/1.1
08:40:28.054556 IP 192.168.1.1.http > 192.168.1.254.21014:
length 0
```

Alternatively, you can use the built-in Wireshark capture tool in EVE-NG to monitor HTTP traffic directly. Right click one **SRV2 > Capture** choose interface **e0**



SRV2_e0

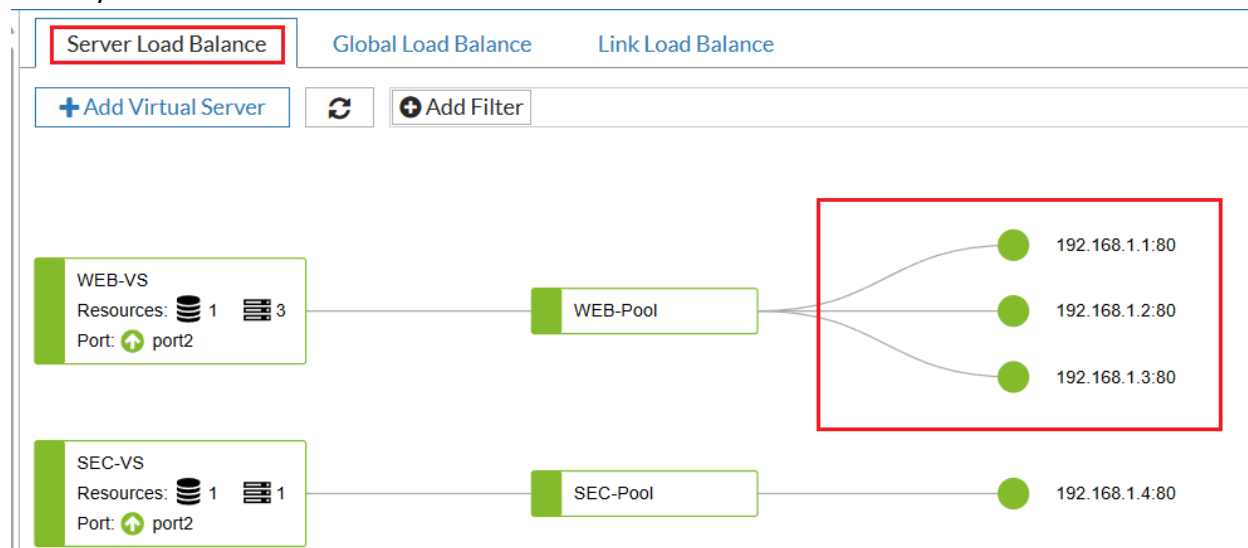
No.	Time	Source	Destination	Protocol	Length	Info
20	8.161667512	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
29	13.163735487	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
31	13.164134782	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
43	18.166893444	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
45	18.167601192	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
54	23.169930238	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
56	23.170426444	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
66	28.173507434	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
68	28.174120133	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
77	33.176001701	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
79	33.176473652	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
89	38.178738709	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
91	38.179139509	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
100	43.182397243	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
102	43.182805958	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
115	48.185505904	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
117	48.185883076	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)



FortiADC_port3

No.	Time	Source	Destination	Protocol	Length	Info
11	1.059652011	192.168.1.254	192.168.1.1	HTTP	194	GET /index.html HTTP/1.1
12	1.059712913	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
13	1.059784072	192.168.1.254	192.168.1.3	HTTP	194	GET /index.html HTTP/1.1
17	1.060199960	192.168.1.1	192.168.1.254	HTTP	528	HTTP/1.1 200 OK (text/html)
18	1.060206434	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
21	1.060407009	192.168.1.3	192.168.1.254	HTTP	529	HTTP/1.1 200 OK (text/html)
41	6.062113966	192.168.1.254	192.168.1.1	HTTP	194	GET /index.html HTTP/1.1
42	6.062134251	192.168.1.254	192.168.1.2	HTTP	194	GET /index.html HTTP/1.1
43	6.062137392	192.168.1.254	192.168.1.3	HTTP	194	GET /index.html HTTP/1.1
47	6.062939295	192.168.1.2	192.168.1.254	HTTP	530	HTTP/1.1 200 OK (text/html)
48	6.062978845	192.168.1.1	192.168.1.254	HTTP	528	HTTP/1.1 200 OK (text/html)
49	6.063048494	192.168.1.3	192.168.1.254	HTTP	529	HTTP/1.1 200 OK (text/html)

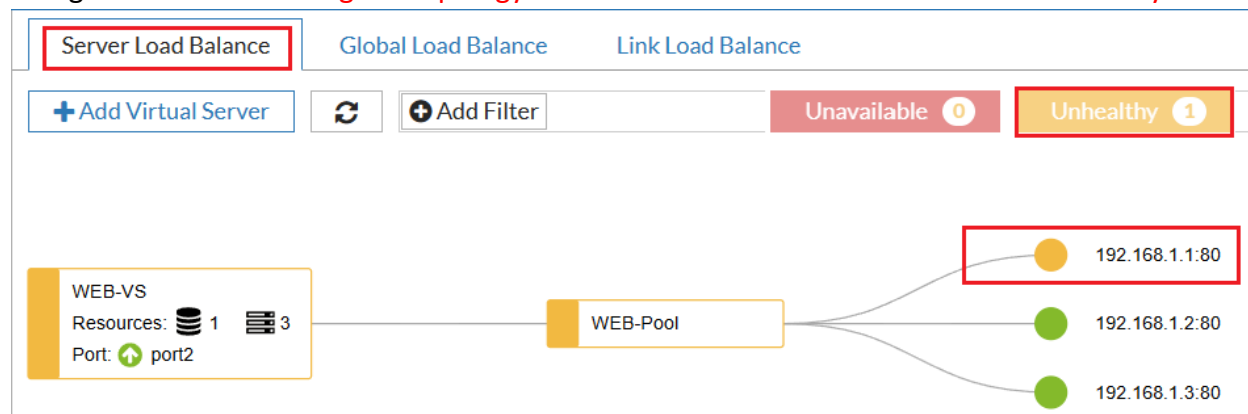
Navigate to **FortiView>Logical Topology>Server Load Balance** all the servers are up green and healthy.



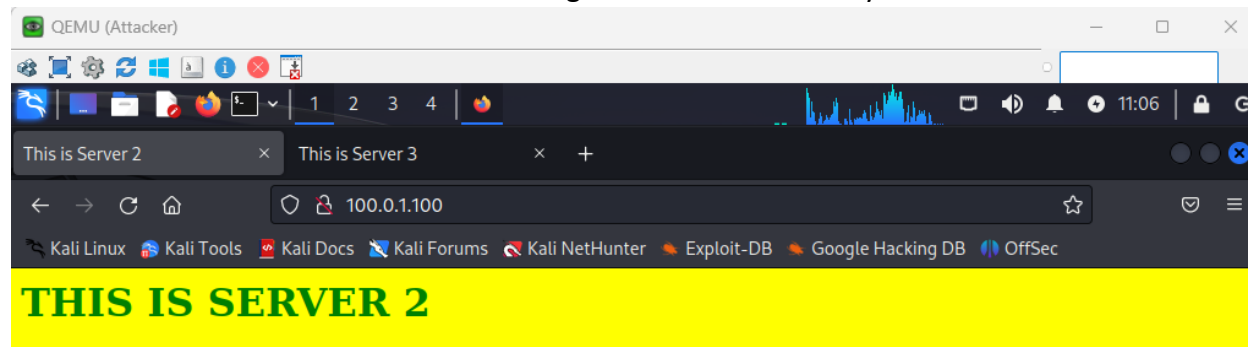
To test the HTTP-Based Health check let's stop one of the Server in this case **SRV1**.

root@SRV1:~# shutdown -h now

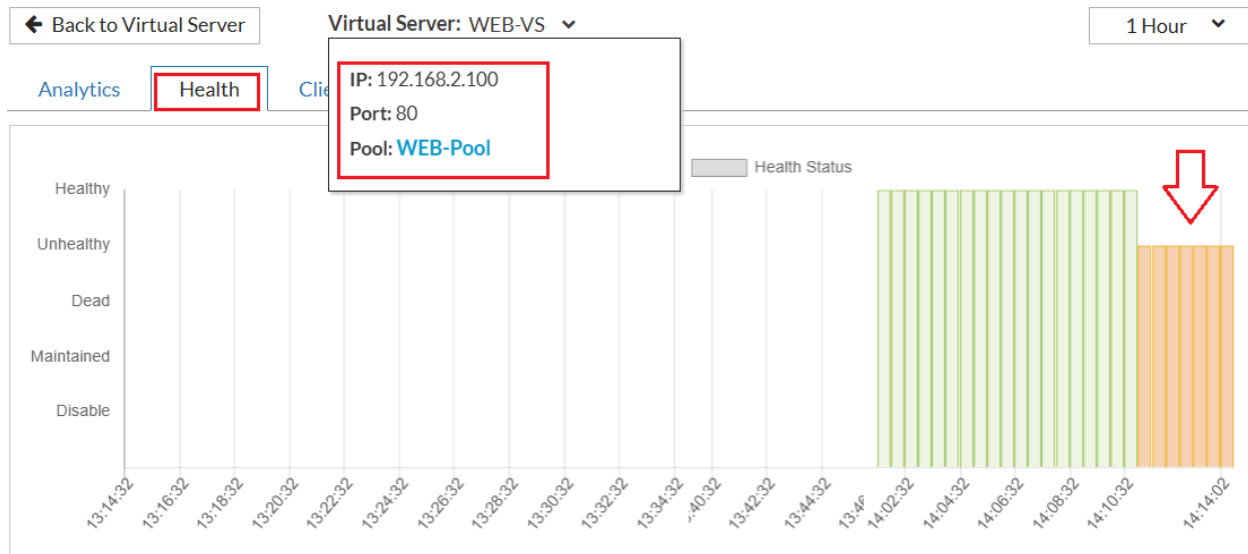
Navigate to **FortiView>Logical Topology>Server Load Balance** this time **SRV1** is unhealthy



Based on the health FortiADC is not sending traffic to **Server 1** only to **SERVER 2** and **SERVER 3**



Navigate to **FortiView>Virtual Server** click on the name of Virtual Server in this case **WEB-VS** after that click on **Health** tab to check Health Status the organ color show is unhealthy.



Let's start **SERV1**, but this time, stop only the **Web Services** running on it. This approach allows us to simulate a scenario where the server itself is operational but the application or web service is unavailable.

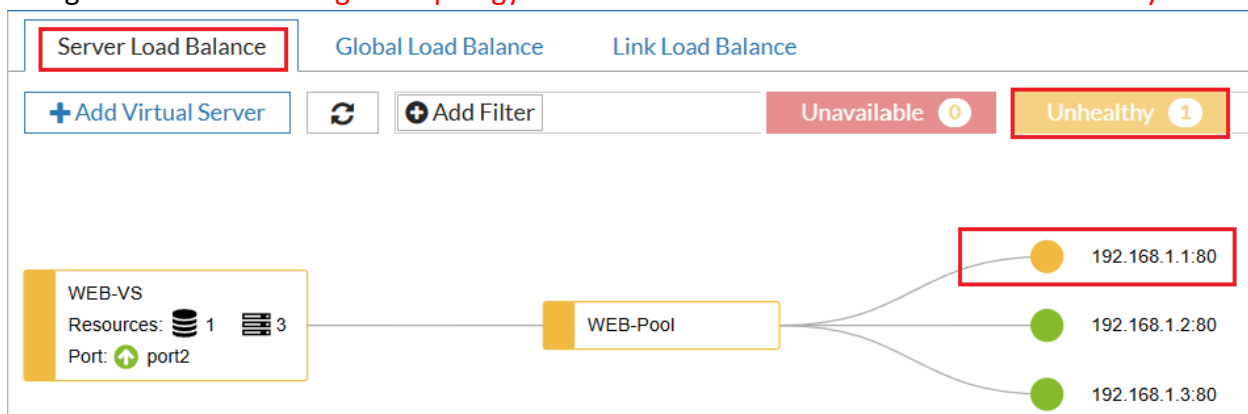
root@SRV1:~# service apache2 stop

```

root@SRV1:~# service apache2 stop
root@SRV1:~#

```

Navigate to **FortiView>Logical Topology>Server Load Balance** this time **SERV1** is unhealthy



root@SRV1:~# service apache2 start

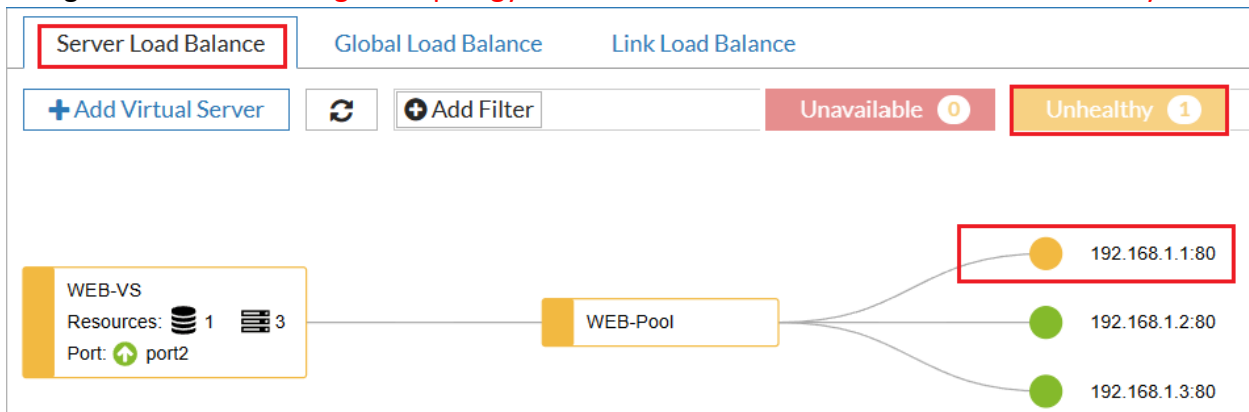
The server is operational, and the port is accessible; however, the resources are currently unavailable.

```
root@SRV1:~# cd /var/www/html/
root@SRV1:/var/www/html# ls
index.html
root@SRV1:/var/www/html# rm index.html
```

Go to SRV1 login with username & password root/root type **tcpdump tcp**. This time the resources **404 not Found**.

```
root@SRV1:~# tcpdump tcp
tcpdump: verbose output suppressed, use -v or -vv for full pr
listening on ens3, link-type EN10MB (Ethernet), capture size
08:49:38.451522 IP 192.168.1.254.60548 > 192.168.1.1.http: F
38495 ecr 0,nop,wscale 9], length 0
08:49:38.451626 IP 192.168.1.1.http > 192.168.1.254.60548: F
ckOK,Ts val 199892 ecr 1789338495,nop,wscale 7], length 0
08:49:38.452442 IP 192.168.1.254.60548 > 192.168.1.1.http: F
length 0
08:49:38.452446 IP 192.168.1.254.60548 > 192.168.1.1.http: F
r 199892], length 128: HTTP: GET /index.html HTTP/1.1
08:49:38.457283 IP 192.168.1.1.http > 192.168.1.254.60548: F
, length 0
08:49:38.460779 IP 192.168.1.1.http > 192.168.1.254.60548: F
1789338496], length 502: HTTP: HTTP/1.1 404 Not Found
08:49:38.463274 IP 192.168.1.254.60548 > 192.168.1.1.http: F
length 0
```

Navigate to **FortiView>Logical Topology>Server Load Balance** this time **SRV1** is **unhealthy**



Configure Back the Server 1

```
#!/bin/bash
sudo cat > /var/www/html/index.html << EOF
<html>
<head>
<title>This is Server 1</title>
</head>
<body style="background-color:yellow;">
<h1 style="color:red;">THIS IS SERVER 1</h1>
</body>
</html>
EOF
```

Navigate to **FortiView>Logical Topology>Server Load Balance** all the servers are up green and healthy.

